FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VIII

JEE (Advanced)-2020

PAPER – 2

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. B

Sol. $a_m = 4\,a_M$

if $a_M = a$

$a_m = 4\,a$

$= \sqrt{a^2 + (4a)^2 + 2(a)(4a)\cos(180-\alpha)}$

$= a\sqrt{17 - 8\cos\alpha}$

2. B

Sol. Virtual power = 0

$-Ta_1 - \frac{T}{2}a_2 - \frac{T}{4}a_3 \ldots = 0$

$a_1 + \frac{a_2}{2} + \frac{a_3}{4} + \ldots = 0$

Newton second law

$mg - T = ma_1$

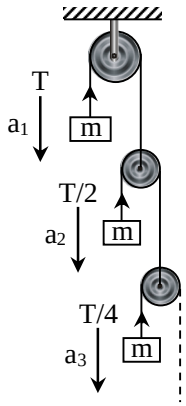
$a_1 = g - \frac{T}{m}$

$mg - \frac{T}{2} = ma_2$

$a_2 = g - \frac{T}{2m}$

$a_3 = g - \frac{T}{4m}$

put value of $a_1 a_2$ ……. (1)



$g + \frac{g}{2} + \frac{g}{4} + \ldots \infty = \frac{T}{m}\left[1 + \frac{1}{4} + \frac{1}{16} + \ldots\infty\right]$

$T = \frac{3mg}{2}$ $\therefore a_1 = g - 1.5\,g = -\frac{g}{2}$

3. C

Sol. $\Delta Q = \frac{dh}{dt} = \left(\frac{27}{L}\right)K \times \frac{1}{9} = \frac{3K}{L} J/\sec$

First law of thermodynamic for the cylinder

$\frac{dQ}{dt} = \frac{dW}{dt} + \frac{dU}{dt}$

as T = constant, $\frac{dU}{dt} = 0$

$\therefore \frac{3V}{L} = P\frac{dV}{dt}$

$P \times \frac{L}{2}A = 0.5 \times R \times 300$

$P = \frac{300R}{LA}$

$\frac{3K}{L} = \frac{300R}{L} 9 \times \frac{dV}{dt}$

$\frac{3K}{L} = \frac{300R}{L} \times 9 \times A\frac{dL}{dt}$

$\frac{dL}{dt} = V = \frac{K}{100R}$

4. B

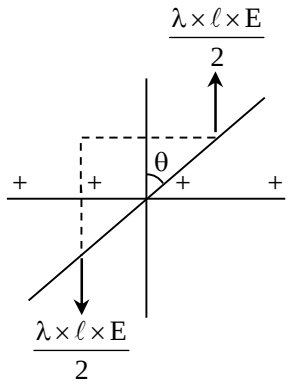
Sol. $\tau = I\alpha$

$\frac{\lambda \ell \times E}{2} \times \frac{\ell}{2}\sin\theta = \frac{M\ell^2}{12} \times \alpha$

$\alpha = -\frac{3\lambda E}{M}\theta$

$\omega = \sqrt{\frac{3\lambda E}{m}}$

$T = \frac{2\pi\sqrt{m}}{\sqrt{3\lambda E}} = 2\pi\sqrt{\frac{m \times 2 \in_0}{3\lambda\sigma}}$



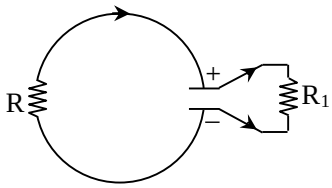
5. A

Sol. $e = -\pi a^2 \frac{dB}{dt} = \pi a^2 k$

$a = Ce = q\pi a^2 k$

B decreasing

$\therefore$ induced current is clockwise.

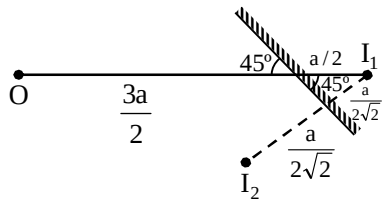


6. C

Sol. Through lens, the image of 'O' will be formed at distance a behind the lens on the principle axis itself and it will behave as virtual object for plane mirror and $\frac{a}{2}$ behind the mirror

So, the separation between O and I_2 is

$$\sqrt{\left(\frac{3a}{2}\right)^2+\left(\frac{a}{2}\right)^2}=\frac{\sqrt{10}a}{2}$$



7. ABCD

Sol. Equilibrium of piston gives

$PS = kx_0$ or $P = \frac{kx_0}{S}$

Since, the chamber is thermally insulated

$\Delta Q = 0$

$\therefore$ Elastic potential energy of spring = work done by gas

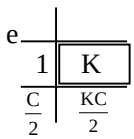
or work done by gas = $\frac{1}{2}kx_0^2$

This work is done in the expense of internal energy of the gas. Therefore, internal energy of the gas is decreased by $\frac{1}{2}kx_0^2$.

Internal energy of an ideal gas depends on its temperature only. Internal energy of the gas is decreasing. Therefore, temperature of the gas will decrease.

8. ABC

Sol. V remain same as capacitor is connected with battery charge on C = C'V



$C_{eq} = \frac{C}{2}+\frac{KC}{2} = \frac{(K+1)C}{2}$

Charge on C = $\frac{(K+1)}{2}CV$

energy stored in C when slab is

half filled = $\frac{1}{2}C'V^2 = \frac{1}{2}\frac{(K+1)C}{2}V^2$

Total work done = KE – 0

$W_{\text{by battery}} + T \times \frac{\ell}{2} + W \text{ by } F_{\text{electric}} = KE$

$W_{\text{by battery}} + T \times \frac{\ell}{2} - \left[\frac{1}{2}C'V^2 - \frac{1}{2}CV^2\right] = KE$

$W_{\text{by battery}} + T \times \frac{\ell}{2} - \frac{1}{2}V^2\left[\frac{(K+1)}{2}C - KC\right]$

$W_{\text{by battery}} + T \times \frac{\ell}{2} - \frac{1}{2}V^2\frac{[1-K]C}{2} = KE$

$W_{\text{by battery}} = (Q_t - Q_i) \times V$

$= \left[\frac{(K+1)CV}{2} - CV\right]V = \frac{[K-1]}{2}CV^2$

$Mg - T = Ma$

$T = Ma$

$T = \frac{mg}{2}$

$\frac{[1-K]CV^2}{2} - \frac{1}{4}CV^2[K-1] + \frac{mg\ell}{4} = KE$

$$KE = \frac{mg\ell}{4} - \frac{CV^2(K-1)}{8}$$

9. D

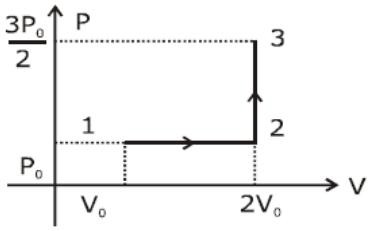
Sol. $I \to Q;\ II \to R;\ III \to S;\ IV \to U$

$$X = \frac{3}{2}R\ \ell n\frac{T}{T_A} + R\ell n\frac{V}{V_A}$$

$$dx = \frac{3}{2}R\frac{1}{T}dT + \frac{R}{V}dv$$

$$d\theta = Tdx$$

$$d\theta = \frac{3}{2}RdT + \frac{RT}{V}dv$$



$$\int dQ = \int \frac{3}{2}RdT + \int_{V_0}^{2V_0}\int \frac{RT}{V}dv$$

(iv) $$Q_{1\to2} = \int \frac{3}{2}RdT + \int_{V_0}^{2V_0}\frac{RT}{V}dv$$

$$Q = \frac{3}{2}R\int_{\frac{2T_0}{3}}^{T_0} dT + \int pdv$$

$$Q = \frac{3}{2}R\frac{T_0}{3} + P_0V_0$$

$$Q_{1\to2} = \frac{RT_0}{2} + \frac{RT_0}{3} = \frac{5}{6}RT_0 \quad \{\because P_0V_0 = \frac{RT_0}{3}$$

(iii) $$Q_{2\to3} = \int_{\frac{2T_0}{3}}^{T_0}\frac{3}{2}RdT + 0$$

$$Q_{2\to3} = \frac{RT_0}{2}$$

$$Q_{1\to3} = \frac{5}{6}RT_0 + \frac{RT_0}{2} = \frac{4}{3}RT_0$$

(ii) $$\Delta u = \frac{f}{2}nR\Delta T = \frac{3}{2}R\frac{2T_0}{3} = RT_0$$

(1) $dq = dw + du$

$dw = dq - du$

$$w = \frac{4}{3}RT_0 - RT_0$$

$$W = \frac{RT_0}{3}$$

10. A

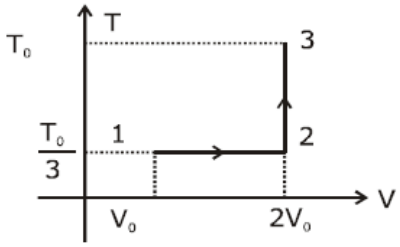
Sol. I→P; II→R; III→T; IV→P

$$X = \frac{3}{2}R\ \ell n\frac{T}{T_A} + R\ell n\frac{V}{V_A}$$

$$dx = \frac{3}{2}R\frac{1}{T}dt + \frac{R}{V}dv$$

$$dQ = Tdx$$

$$dQ = \frac{3}{2}RdT + \frac{RT}{V}dv$$



(iv) $\int dQ = \int \frac{3}{2}Rdt + \int \frac{RT}{V}dv$

$$Q = 0 + \frac{RT_0}{3}\left(\ell nv\right)_{v_0}^{2v_0}$$

$$Q_{1\to2} = \frac{RT_0}{3}\ell n2$$

(iii) $Q_{2\to3} = \int_{\frac{T_0}{3}}^{T_0}\frac{3}{2}Rdt + 0 = \frac{3}{2}Rto\frac{2}{3} = RT_0$

$$Q_{1\to3} = Q_{1\to2} + Q_{2\to3}$$

$$= \frac{RT_0}{3}(3 + \ell n2)$$

(ii) Change in internal energy

$$\Delta u = \frac{f}{2}nR\Delta t$$

$$\Delta u = \frac{3}{2}R\left(T_0 - \frac{T_0}{3}\right)$$

$$\Delta u = RT_0$$

(i) $dq = dw + \Delta u$

$dw = dq - \Delta u$

$$= \frac{1}{3}RT_0\ell n2$$

11. B

Sol. Fundamental frequency $f_0 = f_0 = \frac{1}{2L_0}\sqrt{\frac{T_0}{u}} \rightarrow (1)$

Ist string $f_1 = \frac{1}{2\ell_0}\sqrt{\frac{T_0}{\mu}}$

$f_1 = f_0$

IInd string $f_2 = \frac{1}{2L_0}\sqrt{\frac{T_0}{2\mu}}$

$f_2 = \frac{1}{\sqrt{2}} f_0$

IIIrd string

$f_3 = \frac{1}{2L_0}\sqrt{\frac{T_0}{3\mu}}$

$f_3 = \frac{1}{\sqrt{3}} f_0$

IVth string

$f_4 = \frac{1}{2L_0}\sqrt{\frac{T_0}{4\mu}}$

$f_4 = \frac{1}{2} f_0$

12. B

Sol. I→P; II→Q;III→T; IV→U

fundamental frequency

$f_0 = \frac{1}{2\ell}\sqrt{\frac{T_0}{\mu}}$

Ist string → 1 st harmonic

$f_1 = \frac{1`}{2\ell}\sqrt{\frac{T_0}{\mu}}$

$f_1 = f_0$

so $\Rightarrow T_1 = T_0$

IInd string → IIIrd Harmonic

$\ell = \frac{3\ell_0}{2}$

$f_2 = 3 \times \frac{1}{2 \times \frac{3\ell_0}{2}}\sqrt{\frac{T_2}{2\mu}} = \frac{1}{\ell_0}\sqrt{\frac{T_2}{2\mu}}$

$\because f_2 = f_0$

So $T_2 = \frac{T_0}{2}$

IIIrd string

$\ell = \frac{5\ell_0}{4}$ → Fifth harmonic

$$f_3 = 5 \times \frac{1}{2 \times \frac{5\ell_0}{4}} \times \sqrt{\frac{T_3}{3\mu}}$$

$$f_3 = \frac{2}{\ell_0}\sqrt{\frac{T_3}{3\mu}}$$

$\because f_3 = f_0$

$$\frac{2}{\ell_0}\sqrt{\frac{T_3}{3\mu}} = \frac{1}{2}\sqrt{\frac{T_0}{\mu}}$$

$T_3 = \frac{3T_0}{16}$

IVth streing → IV harmonic

$$f_4 = 14 \times \frac{1}{\frac{2 \times 7\ell_0}{4}} \sqrt{\frac{T_4}{4\mu}}$$

$$f_4 = \frac{4}{\mu_0}\sqrt{\frac{T}{4\mu}}$$

$f_4 = f_0$

$T_4 = \frac{T_0}{16}$

SECTION – C

13. 00002.00

Sol. Magnetic field ($\vec{B}$) at the origin = Magnetic field due to semicircle KLM + magnetic field due to other semicircle KNM.

Therefore, $\vec{B} = \frac{\mu_0 I}{4R}(-\hat{i}) + \frac{\mu_0 I}{4R}(\hat{j})$

$\Rightarrow \quad \vec{B} = \frac{\mu_0 I}{4R}\hat{i} + \frac{\mu_0 I}{4R}\hat{j} = \frac{\mu_0 I}{4R}(-\hat{i} + \hat{j})$

[$\vec{B}$ due to a circular current carrying loop is $\frac{\mu_0 I}{2R}$

$\therefore$ For semicircle it is half]

Therefore, magnetic force acting on the particle.

$$\vec{F} = q(\vec{v} \times \vec{B}) = q\left\{(-v_0\hat{i}) \times (-\hat{i} + \hat{j}) \times \frac{\mu_0 I}{4R}\right\}$$

$$= \frac{-\mu_0 q v_0 I}{4R}\,\hat{k}$$

$$\therefore |\vec{F}| = \frac{\mu_0 q v_0 I}{4R} = \frac{8R}{4R} = 2 \text{ units}$$

14. 00001.00

Sol. $\mu = eV$

$$F = \frac{d\mu}{dr}$$

$$\Rightarrow \frac{d}{dr}\,(v_0) \log\left(\frac{r}{r_0}\right)$$

$$F = \frac{v_0}{r} \cdot r_0 = \frac{mv^2}{r}$$

v = constant

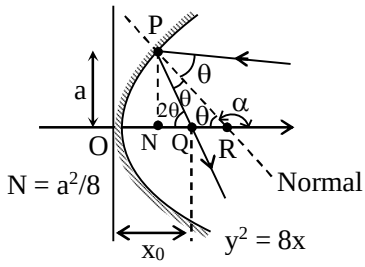
$$mvr = \frac{nh}{2\pi}$$

$r \propto n$

15. 00002.00

Sol. In ΔPNQ

$$\tan 2\theta = \frac{a}{\left(x_0 - \frac{a^2}{8}\right)} \qquad \ldots(i)$$



$y^2 = 8x$

$$\frac{dy}{dx} = \frac{4}{y}$$

at P $y = a$

$$\frac{dy}{dx} = \frac{4}{a} = \tan(90^\circ - \theta)$$

or $\boxed{\tan\theta = \frac{a}{4}}$ …(ii)

eq.(i) and (ii)

$x_0 = 2$

16. 00003.00

Sol. We have

$$E = 13.6\, Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \text{ (in eV)}$$

$$10.2 + 17 = 13.6\, Z^2 \left(\frac{1}{2^2} - \frac{1}{n^2}\right) \qquad \ldots(i)$$

and

$$4.25 + 5.95 = 13.6\, Z^2 \left(\frac{1}{3^2} - \frac{1}{n^2}\right) \qquad \ldots(ii)$$

from (i) & (ii)

$\boxed{n = 6}$ and $\boxed{Z = 3}$

17. 00002.00

Sol. At A

$V_A = 0$

Ie

$V_{due\ to\ q} + V_{induced} = 0$

$$\frac{1}{4\pi\varepsilon_0}\frac{q}{(3.5R)} + V_{induced} = 0$$

$$V_{induced} = \frac{28\pi\varepsilon_0 R}{14\pi\varepsilon_0 R} = 2$$

18. 00002.00

Sol. 10 t = 2T

T = 5t

Block A will lose the contact at $5t_1 = m_A g$,

$t_1 = 2$ sec.

Block B will lose the contact at $5t_2 = m_B g$

$t_2 = 4$ sec

For Block A between the two events,

$$5t - m_A g = m_A \frac{dv}{dt}$$

$$\int_2^4 (5t - 10)\,dt = \int_0^v dv$$

v = 10 m/sec

Chemistry

PART – II

SECTION – A

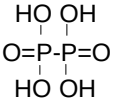
19. ABCD

Sol. Let, $r = k[A]^x[B]^y$

from the given data, $x = 1$ & $y = 1$

20. ACD

Sol. The structural formula of $H_4P_2O_6$(Hypophosphoric acid) is



21. AB

Sol. The five Bravais lattices in 2-D are: oblique (monoclinic), rectangular (orthorhombic), cantered rectangular (orthorhombic), hexagonal, square (tetragonal)

A cube has nine plane of symmetry. Three of the planes run parallel to the faces of the cube, and the other six run diagonally from one edge to the opposite edge.

For monoclinic system, $a \neq b \neq c$ and $\alpha = \gamma = 90^0, \beta \neq 120^0$

The packing fraction of BCC is 68%

22. ABCD

Sol. Apply the formula to get answers

23. BD

Sol. (i) $\text{Trans} - 2 - \text{butene} \xrightarrow{Br_2/CHCl_3} 2,3 - \text{Dibromobutane(Meso compounds)}$

(ii) $\text{Cis} - 2 - \text{butene} \xrightarrow{Br_2/CHCl_3} 2,3 - \text{Dibromobutane(Pair of enantiomers)}$

Bromination proceeds through trans – addition in both the reactions

24. BD

Sol. $K_4[Fe(CN)_6] + H_2SO_4 + H_2O \longrightarrow K_2SO_4 + CO + FeSO_4 + (NH_4)_2SO_4$

$CuSO_4 + NH_3 \longrightarrow [Cu(NH_3)_4]SO_4$

25. ACD

Sol. The ratio of $\frac{[X]}{[Y]}$ is a constant equal to $\frac{k_1}{k_2}$, only when order is same for both parallel reactions.

The overall energy of activation, $E = \frac{k_1E_1 + k_2E_2}{k_1 + k_2}$.

For parallel first order reaction, $\ln t_{1/2} = \ln\left(\frac{0.693}{A}\right) + \frac{E}{R}\left(\frac{1}{T}\right)$.

26. BC

Sol. $Zn^{2+} + 2e^- \longrightarrow Zn(s)$ (1 F $\longrightarrow$ 65/2 g)

$Mg^{2+} + 2e^- \longrightarrow Mg(s)$ (1 F $\longrightarrow$ 24/2 g)

$Al^{3+} + 3e^- \longrightarrow Al(s)$ (1 F $\longrightarrow$ 27/3 g)

$10H^- \longrightarrow H_2O + \frac{1}{2}O_2 + Ze^-$ (1 F $\longrightarrow$ 8 g)

27. A

28. D

Sol. **(for Q. 27 & 28)**

$Q = mS\Delta T$

$$\Delta T \propto \frac{Q}{m}$$

$$\frac{\Delta T_1}{\Delta T_2} = \frac{(Q/m)_1}{(Q/m)_2}$$

$$HA + BOH \longrightarrow BA + H_2O$$

Milliequivalent 10 10

– –

When $\frac{Q}{m} = \frac{10\times10^{-3}\Delta Hneut}{20\times10^{-3}} = \frac{\Delta Hneut}{2}, \Delta T = 5°C$

In (A) when $\frac{Q}{M} = \frac{100\Delta Hneu}{200} = \frac{\Delta Hneut}{2}, \Delta T = 5°C$

In (B) $\frac{Q}{M} = \frac{10\Delta Hneut}{30} = \frac{\Delta Hneut}{3}$, $\Delta T' = ?$

$$\frac{\Delta T'}{5} = \frac{\frac{\Delta Hneut}{3}}{\frac{\Delta Hneut}{2}} = \frac{2}{3} \quad \Delta T' = \frac{10}{3} = 3.3°C$$

In (C) $\frac{Q}{M} = \frac{20\Delta Hneu}{50} = \frac{2}{5}\Delta Hneu$ $\Delta T'' = ?$ $\frac{\Delta T''}{5} = \frac{\frac{2}{5}}{\frac{1}{2}} = \frac{4}{5}$ $\Delta T'' = 4°C$

In (D) $\frac{\Delta T}{5} = \frac{\frac{50\times\Delta Hneu}{200}}{\Delta Hneu/2}$ $\Delta T = 2.5°C$

[Ans:- (I) – P ; (II) – R ; (III) – Q ; (IV) – T]

29. A

30. C

Sol. **(for Q.29 & 30)**

In (A) $pH = 7 + \frac{1}{2}pKa + \frac{1}{2}\log C = 7 + \frac{5}{2} + \frac{1}{2}\log\frac{5}{10} = 7 + 2.5 + \frac{1}{2}(-0.3010) = 9.34$

In (B) Since acid is left along with salt, we have a buffer $pH = 5 + \log\frac{10}{5} < 7$

In (C) $pOH = 5 + \log\frac{5}{10} = 5 - 0.3010 = 4.699$

[Ans:- (I) – P, S ; (II) – Q, R ; (III) – P, R ; (IV) – P]

SECTION – C

31. 00004.40

Sol. The balanced chemical equation is

$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$

1 mol 2 mol 1 mol

16 gm 2×22.4 L 22.4 L

44 gm

No. of moles of $CH_4 = \frac{8}{16} = 0.5\,\text{mol}$

No. of moles of $O_2 = \frac{4.48\text{L}}{22.4\text{L}} = 0.2\text{mol}$

Now since 1 mole of CH_4 requires 2 mol (i.e. 44.8 L) of O_2 for complete combustion. But the given moles of O_2 is only 0.2 mol. So, O_2 is the limiting reagent.

Again, since 2 moles of O_2 reacts with 1 mol of CH_4 to give 22.4 L of CO_2 at STP.

So 0.2 mole of O_2 will react with 0.1 mol of CH_4 to give 2.24 L of CO_2.

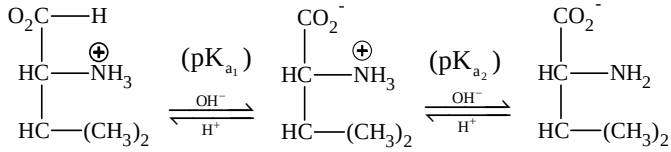
Wt. of CO_2 produced = 0.1 mol $\times$ 44 = 4.4 gms of CO_2

32. 00006.00

Sol. The isoelectric point (pH_i) is the pH at which the amino acid exists only as a dipolar ion with net charge zero.

At isoelectric point, for a neutral amino acid, $\text{pH}_\text{i} = \frac{\left(\text{pK}_{a_1} + \text{pK}_{a_2}\right)}{2}$

The dissociation of cationic form of valine can be represented as

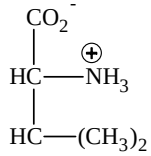


Netcharge (+1) (0) (-1)

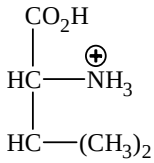
The species with zero net charge exists between species with (+1) and (–1) net charges.

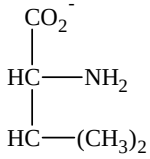
$$\text{pH}_\text{i} = \frac{\left(\text{pK}_{a_1} + \text{pK}_{a_2}\right)}{2} = \frac{9.69 + 2.31}{2} = 6$$

When the pH of the solution equals to pH_i, the structure of valine is



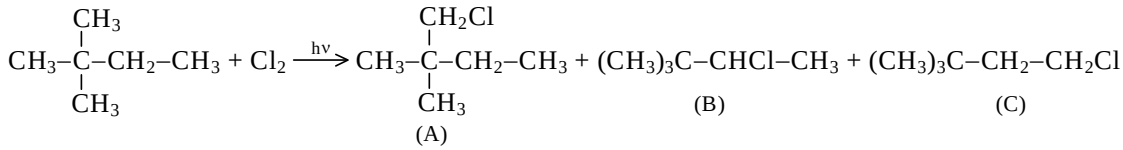
When the pH of the solution is two, the structure of valine is



When the pH of the solution is 12, the structure of valine is 

33. 00046.00

Sol. Photochlorination of 2,2–dimethylbutane gives three products as shown below



(A) (B) (C)

	Reactivity factor	×	Probability factor	=	Number of parts
(A) :	1	×	9	=	9
(B) :	3.8	×	2	=	7.6
(C) :	1	×	3	=	3
			Total number of parts	=	19.6

The compound is the major product. Its percentage is given by

Percentage of (A) = $\frac{9\times100}{19.6}$ = 45.9 $\simeq$ **46%**.

34. 00009.01

Sol. $\Delta T_f = T_f^0 - T_f = K\cdot m$

5.5°C – 3.5°C = 5.12 K/m × m

$$m = \frac{2}{5.12} = 0.39m$$

$\therefore$ Mass of iodine needed for 1000g of benzene = m × molecular mass of iodine I_2

= 0.39 mol/kg × 254 g/mol = 99.06 g/kg

$\therefore$ 1000g + 99.06g solution contains 99.06g I_2

100g solution contains $\frac{99.06g\times100}{1099.06g} = 9.01\%$

35. 00000.25

Sol. Dipole moment for a diatomic molecule (μ) = e × d

$\therefore$ electronic charge = $\frac{\mu}{d} = \frac{1.2\times10^{-18}\,esu.cm}{1.0\times10^{-8}\,cm}$ = 1.2 × 10^{-10} esu

$\therefore$ fraction of electronic charge = $\frac{1.2\times10^{-10}}{4.8\times10^{-10}}$ = 0.25

36. 00000.03

Sol. With the help of calculation of free energy

$\Delta G_1^0 = -nF\varepsilon^0 = -(1)(F)(0.771V) = -0.771V$

$\Delta G_2^0 = -(2)(F)(0.44) = 0.88FV$

$\Delta G_3^0 = -(3)(F)(\varepsilon^0) = -3Fe^0V$

The free energy change for the unknown process can be obtained as

$\Delta G_3^0 = \Delta G_1^0 + \Delta G_2^0$

$-3F\varepsilon^0 = -0.771F + 0.88F = 0.109F$

or $\varepsilon^0 = -\frac{0.109}{3} = -0.036V$

Mathematics

PART – III

SECTION – A

37. ACD

Sol. $f(x) = \cos\pi(|x| + [x])$

$f(0) = 1.f(0-0) = \cos\pi(-1) = -1$

$f(1/2) = \cos\pi\left(\frac{1}{2}\right) = 0,\ f\left(\frac{1}{2}, 0\right)\cos\pi\left(\frac{1}{2}\right) = 0$

$f\left(\frac{1}{2} + 0\right) = \cos\frac{\pi}{2} = 0$

for $x \in (0,1) f(x) = \cos\pi x$

for $x \in (-1,0) f(x) = \cos\pi(-x-1) = -\cos\pi x$

38. AC

Sol. $OP = 5\sqrt{2}\sec\theta$

$OP_1 = 5\sqrt{2}\,\text{cosec}\theta$

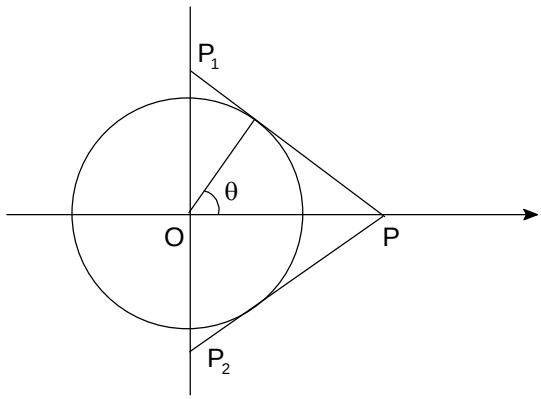
$\Delta PP_1P_2 = \frac{100}{\sin 2\theta}$

$(\Delta | PP_1P_2)_{min} = 100$

$\Rightarrow \theta = \frac{\pi}{4}$

$\Rightarrow OP = 10$

$\Rightarrow P = (10, 0), (-10, 0)$



39. AB

Sol. Let P(2, -1) goes 2 units along x + y = 1 upto A and 5 units along x – 2y = 4 up to B

Slope of PA = -1 = tan 135^0, slope of PB = $\frac{1}{2} = \tan\theta$

$\therefore \sin\theta = \frac{1}{\sqrt{5}}, \cos\theta = \frac{2}{\sqrt{5}}$

$\therefore A \equiv \left(x_1 + r\cos 135^0,\ y_1 = r\sin 135^0\right) = \left(2 + 2x - \frac{1}{\sqrt{2}}, -1 + 2\times\frac{1}{\sqrt{2}}\right) = \left(2 - \sqrt{2}, \sqrt{2} - 1\right)$

$B \equiv \left(x_1 + r\cos\theta,_1 + r\sin\theta\right) = \left(2 + 5\times\frac{2}{\sqrt{5}},\ -1 + \frac{5}{\sqrt{5}}\right) = \left(2\sqrt{5} + 2,\ \sqrt{5} - 1\right)$

40. AD

Sol. Slope of tangent = $\frac{\frac{-1}{a}}{\frac{1}{b}} = -\frac{b}{a}$

$$\frac{dy}{dx}=\frac{\frac{dy}{dt}}{\frac{dx}{dt}}=\frac{\frac{-4}{t^2}}{4}=\frac{1}{t^2}<0$$

So, $\frac{-b}{a}<0\Rightarrow\frac{b}{a}>0$

41. BC

Sol. $\frac{\sin\frac{\pi}{12}}{\sin x}+\frac{\cos\frac{\pi}{12}}{\cos x}=2$

42. AB

Sol. $|z_1|=|z_2|=1;\ |z_1+z_2|=1$

$\Rightarrow (z_1+z_2)(\bar{z}_1+\bar{z}_2)=1$

$|z_1+z_2|=|z_1||1+e^{i\theta}|=1$

$|z_2-z_1|=|z_1||e^{i\theta}-1|$

43. AB

Sol. $\frac{100}{2}(x_1+x_{100})=-1$

$\frac{50}{2}(x_1+d+x_{100})=1$

44. ABC

Sol. Let $y<x$

$\Rightarrow\frac{y}{x}=\frac{x}{x+y}$

$\Rightarrow R=\frac{1}{1+R}$

45. A

Sol. **I – T; II – P; III – S; IV– Q**.

(I) Let the sides a and b are the roots of $x^2-5x+3=0$, then a + b = 5 and ab = 3

Alos, $\angle C=\frac{\pi}{3}$

$\therefore\frac{1}{2}=\frac{a^2+b^2-c^2}{2ab}\Rightarrow ab=(a+b)^2-2ab-c^2\Rightarrow c=4$

Now, $rR=\frac{abc}{2(a+b+c)}=\frac{3\times4}{2(5+4)}=\frac{2}{3}$

(II) We have $a=\frac{r_1(r_2+r_3)}{\sqrt{r_1r_2+r_2r_3+r_3r_1}}=\frac{8\times36}{24}=12$

(III) Give $1\le a\le10$

Also, for roots of opposite sign $2a^2-7a+3<0$

or $\frac{1}{2} < a < 3 \Rightarrow 1$ or 2

$\therefore$Desired probability $= \frac{2}{10} = \frac{1}{5}$

(IV) (α,β) lies on the director circle of the ellipse i. e. on $x^2 + y^2 = 9$

So, we can assume $\alpha = 3\cos\theta, \beta = 3\sin\theta$.

Thus F = $12\cos\theta + 9\sin\theta = 3(4\cos\theta + 3\sin\theta)$

$\Rightarrow -15 \le F \le 15$

46. D

Sol. **I – T; II – P; III – S; IV– Q**.

(I) Let the sides a and b are the roots of $x^2 - 5x + 3 = 0$, then a + b = 5 and ab = 3

Alos, $\angle C = \frac{\pi}{3}$

$\therefore \frac{1}{2} = \frac{a^2 + b^2 - c^2}{2ab} \Rightarrow ab = (a+b)^2 - 2ab - c^2 \Rightarrow c = 4$

Now, $rR = \frac{abc}{2(a+b+c)} = \frac{3\times 4}{2(5+4)} = \frac{2}{3}$

(II) We have $a = \frac{r_1(r_2 + r_3)}{\sqrt{r_1r_2 + r_2r_3 + r_3r_1}} = \frac{8\times 36}{24} = 12$

(III) Give $1 \le a \le 10$

Also, for roots of opposite sign $2a^2 - 7a + 3 < 0$

or $\frac{1}{2} < a < 3 \Rightarrow 1$ or 2

$\therefore$Desired probability $= \frac{2}{10} = \frac{1}{5}$

(IV) (α,β) lies on the director circle of the ellipse i. e. on $x^2 + y^2 = 9$

So, we can assume $\alpha = 3\cos\theta, \beta = 3\sin\theta$.

Thus F = $12\cos\theta + 9\sin\theta = 3(4\cos\theta + 3\sin\theta)$

$\Rightarrow -15 \le F \le 15$

47. B

Sol. **I – P, R II – P, R; III – Q; IV- Q, R, S, T.**

(I) Let consecutive coefficient be ${}^nC_r$ and ${}^nC_{r+1}$. Then, $\frac{n!}{(n-r)!r!} = \frac{n!}{(n-r-1)!(r+1)!}$

$\Rightarrow \frac{1}{(n-r)(n-r-1)!r!} = \frac{1}{(n-r-1)!(r+1)r!}$

$\Rightarrow r1 = n - r$

$\Rightarrow n = 2r + 1$ hence, n is odd.

(II) $E = (19-4)^n + (19+4)^n$

$2\left[{}^nC_0 19^n + {}^nC_2 19^{n-2}.4^2 + \ldots\ldots\ldots\ldots + {}^nC_{n-1}19.4^{n-1}\right]$ when n is even or

$2\left[{}^nC_0 19^n + {}^nC_2.19^{n-2}.4^2 + \ldots\ldots\ldots + {}^nC_{n-1}19.4^{n-1}\right]$ when n is odd

$\Rightarrow$ E is divisible by19 when n is odd

(III) ${}^{10}C_0\ {}^{20}C_{10} - {}^{10}C_1\ {}^{18}C_{10} + {}^{16}C_2\ {}^{16}C_{10} - .....$

$= \text{Coefficient of } x^{10} \text{ in} \left[{}^{10}C_0 (1+x)^{20} - {}^{20}C_1 \times (1+x)^{18} + {}^{10}C_2 (1+x)^{16} - ..... \right]$

$= \text{Coefficient of } x^{10} \text{ in} \left[{}^{10}C_0 \left((1+x)^2\right)^{10} - {}^{10}C_1 \times \left((1+x)^2\right)^9 + {}^{10}C_2 \left((1+x)^2\right)^8 - ..... \right]$

$= \text{Coefficient of } x^{10} \text{ in} \left[(1+x)^2 - 1 \right]^{10}$

$= \text{Coefficient of } x^{10} \text{ in} \left[2x + x^2 \right]^{10}$

$= 2^{10}$

(IV) $T_r = {}^{14}Cr_{r-1}x^{r-1}; T_{r+1} = {}^{14}C_r x^r; T_{r+2} = {}^{14}C_{r+1}x^{r+1}$

By the given condition, $2^{14}C_r = {}^{14}C_{r-1} + {}^{14}C_{r+1}$

$\Rightarrow 2 = \dfrac{{}^{14}C_{r-1}}{{}^{14}C_r} + \dfrac{{}^{14}C_{r+1}}{{}^{14}C_r}$

$\Rightarrow 2 = \dfrac{r}{14-r+1} + \dfrac{14-(r+1)+1}{r+1}$

$\Rightarrow 2 = \dfrac{r}{15-r} + \dfrac{14-r}{r+1}$

$\Rightarrow r = 9$

48. C

Sol. **I – P, R; II – P, R; III – Q; IV- Q, R, S, T.**

(I) Let consecutive coefficient be ${}^nC_r$ and ${}^nC_{r+1}$. Then, $\dfrac{n!}{(n-r)!r!} = \dfrac{n!}{(n-r-1)!(r+1)!}$

$\Rightarrow \dfrac{1}{(n-r)(n-r-1)!r!} = \dfrac{1}{(n-r-1)!(r+1)r!}$

$\Rightarrow r1 = n-r$

$\Rightarrow n = 2r+1$ hence, n is odd.

(II) $E = (19-4)^n + (19+4)^n$

$2\left[{}^nC_0 19^n + {}^nC_2 19^{n-2}.4^2 + ... + {}^nC_{n-1}19.4^{n-1} \right]$ when n is even or

$2\left[{}^nC_0 19^n + {}^nC_2.19^{n-2}.4^2 + ... + {}^nC_{n-1}19.4^{n-1} \right]$ when n is odd

$\Rightarrow$ E is divisible by19 when n is odd

(III) ${}^{10}C_0\ {}^{20}C_{10} - {}^{10}C_1\ {}^{18}C_{10} + {}^{16}C_2\ {}^{16}C_{10} - .....$

$= \text{Coefficient of } x^{10} \text{ in} \left[{}^{10}C_0 (1+x)^{20} - {}^{20}C_1 \times (1+x)^{18} + {}^{10}C_2 (1+x)^{16} - ..... \right]$

$= \text{Coefficient of } x^{10} \text{ in} \left[{}^{10}C_0 \left((1+x)^2\right)^{10} - {}^{10}C_1 \times \left((1+x)^2\right)^9 + {}^{10}C_2 \left((1+x)^2\right)^8 - ..... \right]$

$= \text{Coefficient of } x^{10} \text{ in} \left[(1+x)^2 - 1 \right]^{10}$

$= \text{Coefficient of } x^{10} \text{ in} \left[2x + x^2 \right]^{10}$

$= 2^{10}$

(IV) $T_r = {}^{14}Cr_{r-1}x^{r-1}; T_{r+1} = {}^{14}C_r x^r; T_{r+2} = {}^{14}C_{r+1}x^{r+1}$

By the given condition, $2^{14}C_r = {}^{14}C_{r-1} + {}^{14}C_{r+1}$

$$\Rightarrow 2 = \frac{{}^{14}C_{r-1}}{{}^{14}C_r} + \frac{{}^{14}C_{r+1}}{{}^{14}C_r}$$

$$\Rightarrow 2 = \frac{r}{14-r+1} + \frac{14-(r+1)+1}{r+1}$$

$$\Rightarrow 2 = \frac{r}{15-r} + \frac{14-r}{r+1}$$

$$\Rightarrow r = 9$$

SECTION – C

49. 00002.00

Sol. $2(1+3+5+7+.....+a)+2(1+3+5+7+.....+b)=2(1+3+5+7+.....+c)$

$$\Rightarrow \left[2+\left(\frac{a+1}{2}-1\right)2\right]\frac{a+1}{2}+\left[2+\left(\frac{b+1}{2}-1\right)2\right]\frac{b+1}{2}\left[2+\left(\frac{c+1}{2}-1\right)2\right]\frac{c+1}{2}$$

$$\Rightarrow (a+1)^2+(b+1)^2=(c+1)^2$$

$\because a > 6$ and $a+b+c=21$

$\therefore a=7, b=5, c=9$

$\therefore a-b=2$

50. 00007.00

Sol. $$p = \frac{{}^3C_1 \times {}^8P_6 + {}^3C_1 \times {}^8P_6 - {}^3P_2 \times {}^7P_5}{2} = \frac{7}{12}$$

51. 00002.00

Sol. $$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$\Rightarrow a_{11}+2a_{12}=a_{11}+a_{21} \quad .....(i)$$

$$a_{11}+a_{12}=2a_{12}=a_{22} \quad .....(ii)$$

$$a_{21}+2a_{22}=2a_{11}+a_{21}$$

$$a_{21}+a_{22}=2a_{12}+a_{22}$$

$\therefore$ From (ii), $a_{11}=a_{22}$

$$a_{21}=2a_{12}$$

$\because a_{11}a_{22}=a_{21}a_{21} \Rightarrow (a_{11})^2=2(a_{12})^2$

$$\Rightarrow a_{11}=\sqrt{2}a_{12}$$

52. 00002.00

Sol. $f''(x)=6x^2-18ax+12a^2=0 \Rightarrow x=a,2a.$

Given $x_2=x_1^2$, so a cannot be negative. If $a>0$ then local minimum occurs at $x=2a$ and local maximum at $x=a$.

Thus $x_1=a$ and $x_2=2a$

$\Rightarrow 2a=a^2 \Rightarrow a=2$ (if $a=0, f(x)$ is always increasing)

53. 00004.00

Sol. Since x, y, z are in A.P., $x + z = 2y$ and hence

$$\frac{x+y}{2y-x}+\frac{y+z}{2y-z}=\frac{x+y}{z}+\frac{y+z}{x}=\frac{y}{x}+\frac{y}{z}+\frac{z}{x}$$

$$=\frac{x}{z}+\frac{x+z}{2z}+\frac{z+z}{2x}+\frac{z}{x}=1+\frac{3}{2}\left(\frac{x}{z}+\frac{z}{x}\right)$$

$$\geq 1+3\left[\text{A.M} \geq \text{G.M}\right]$$

$\Rightarrow$ minimum value of the given expression is 4.

54. 00006.00

Sol. $24300 = 2^2 3^5 5^2$

Number of odd divisors = the number of terms in $\left(1+3+3^2+3^3+3^4+3^5\right)\left(1+5+5^2\right)$

$= 6 + 6 + 6 = 18$

This also includes the factor 1. So the number of proper odd divisors = 17